ASSIGNMENT 2: EXPLORING THE SOLUTION SPACE

The goal of this assignment is to **devise new speculative computing technologies and experiences** by exploring the second stage of the Double Diamond design process.

- **Deliverables:** HMW statement, Mapping of generated ideas, Narrative storyboard
- **Duration:** 2.5 hours

The design issues covered in this assignment are: (1) design process, (2) ideation, (3) storyboarding.

INTRODUCTION: DEVELOP AND DELIVER

In the second assignment, you will focus on the **second diamond of the Double Diamond** design process. You will reuse the problem statement defined in the first assignment and move forward to the second diamond, which includes searching for potential solutions (Develop – divergent thinking) and finding a solution that works (Deliver – convergent thinking).

ASSIGNMENT: DESIGN FOR AN INTERACTIVE TECHNOLOGY & EXPERIENCE

The first step in this assignment is to define a **How Might We (HMW) statement** based on your problem statement. Considering the following problem statement:

> *"Blind children need to be able to participate in collaborative classroom activities to increase their learning performance and social inclusion."*

A HMW statement could be:

> *"How might we support blind children to participate in collaborative classroom activities to increase their learning performance and social inclusion?"*

Notice that we are just **re-framing the problem statement as a question**, which sets you up to explore the solution space.

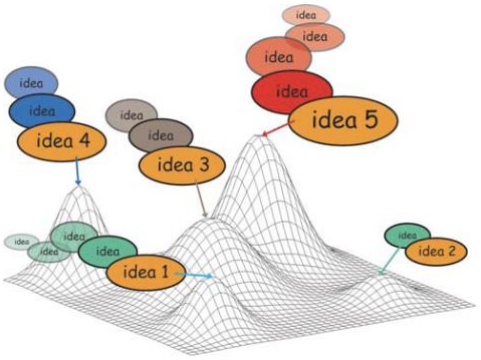
Next, you should **engage in an ideation stage** and be creative! That said, being creative is easier said than done. We do not really understand creativity fully, although researchers are actively trying to understand it[1]. Still, there are some principles and methods that can help you. Be mindful that being creative is a skill and as any skill you need to practice it. Unfortunately, most educational systems do not value creative thinking.

One of the strategies to be creative is quality through quantity. **The best way to have a good idea is to have a lot of ideas**. Your best idea will probably be your 100th and not your first, because it will incorporate all the wonderful things of the first 99 ideas. So, start generating those bad ideas and then making better ones.

To generate many ideas, we recommend start thinking individually and then build on each other's ideas as a team. You can **use time-boxing** brainstorming in both stages; i.e., define a time limit to generate as many ideas as possible (even if they seem weird). Write them down, sketch or use post-it notes (miro boards can be useful). **Do this individually** (each of the team members should generate at least 10 different ideas), then **share** your results with your team, and then **build on the most promising ideas** with another round of time-boxing brainstorming. At this stage, try not to judge the quality of ideas, you want as many as possible.

[1] Wagner, Andreas. Life finds a way: What evolution teaches us about creativity. Basic Books, 2019.

Do not focus just on a single idea, otherwise you risk working on a sub-optimal idea and engage in a "local hill climb" (see Figure below). You want to **simultaneously explore multiple ideas**, searching as much of a design space as possible.



Once you have your idea landscape, it is time to figure out what is wrong with them. How do you figure out what is wrong with your ideas? Externalize. Ideally, you would engage in a prototyping and testing process, where you would create low-fidelity prototypes and test them with real users while iteratively refining them by increasing fidelity.

In this assignment, you will **externalize through** [Narrative Storyboards](), discussing them with your team mates and other students in the studio. Narrative storyboards are useful to illustrate a person using a product to complete a task in a specific context. It is composed by a series of pictures that illustrate a future user experience.



DELIVERABLES

At the end of the second studio, you should **send via email**:

1. **How Might We statement**: the HMW statement should frame the problem as a question.
2. **Generated ideas**: Listing or mapping of all generated ideas.
3. **Narrative storyboard**: (at least one) storyboard illustrating the user experience of a solution you have designed.

GRADING CRITERIA

How Might We statement quality (20%): Is the statement framed as a question? Does it relate to the problem statement?

Generated ideas (40%): What was the number of generated ideas (minimum 30)? Did the team refine some of the ideas?

Narrative storyboard (40%): Does the storyboard show the setting that the user is in? Is it clear the task it needs to complete? Does it show the solution and how it is used? Does it show the benefit for the user?